



STUNTING

Enhancing ASEAN Food Security Through
Comprehensive Food Waste Reduction

INDONESIA

Team Scentia Collective

Ferdi Tan Leonardy | Matthew Owen Gunawan

Mentor by Ade Putera Kemala



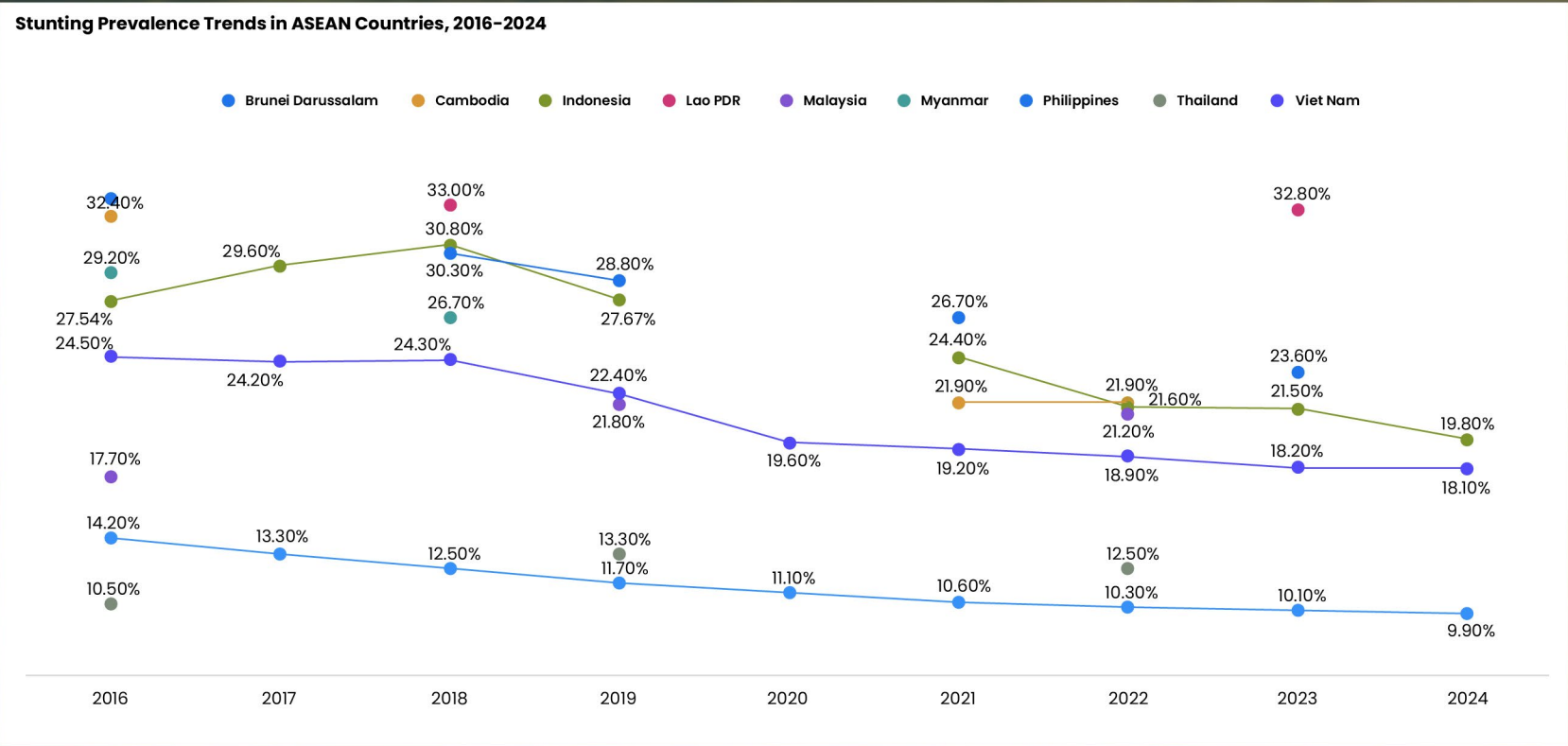
Problem Statement



Though continuous development has been experienced in this region, as per the reports in 2025, 20.3% of the people living in Asia experienced food insecurity to some degree. In addition to these, there are food losses and wastes due to irresponsible consumption and production which pose threats to nutrition and food security. Hence, there is an urgent need to shift towards a circular economy approach.

[https://www.who.int/news/item/21-07-2026-un-report--glob-al-hunger-levels-ease-for-third-consecutive-year-as-regional-disparities-persist](https://www.who.int/news/item/21-07-2026-un-report--global-hunger-levels-ease-for-third-consecutive-year-as-regional-disparities-persist)
<https://unstats.un.org/sdgs/report/2026/goal-12/?utm>

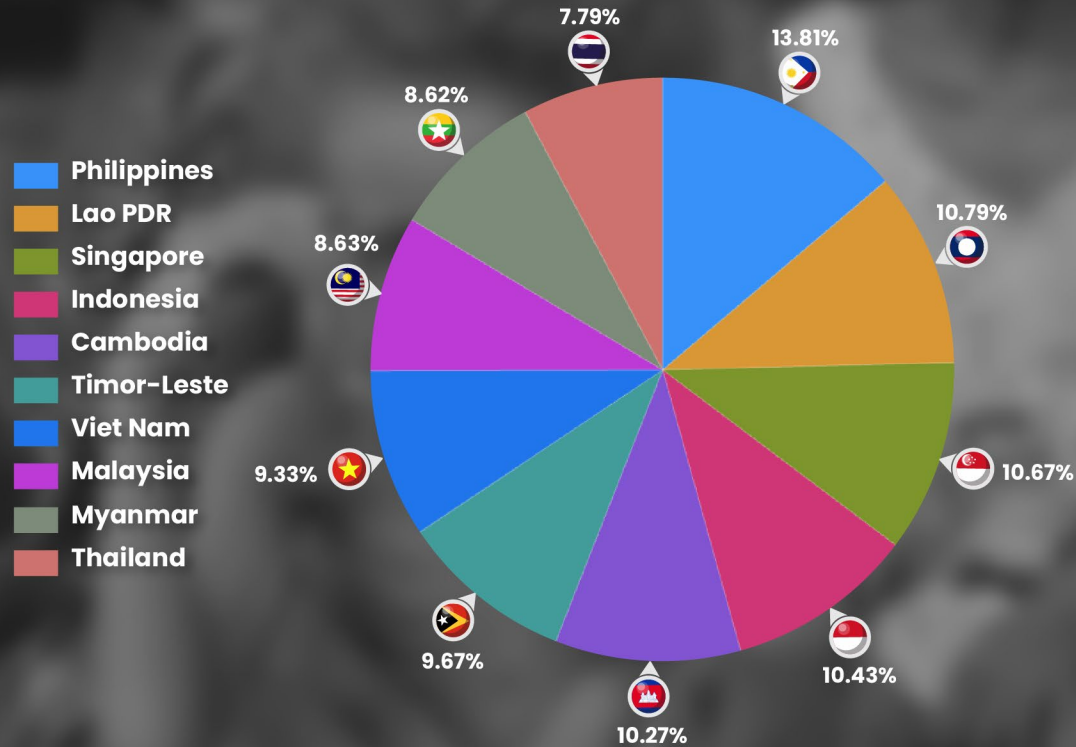
Stunting Pravelance Trends in ASEAN Countries 2016–2024



Stunting rates are declining across several ASEAN countries, however, the prevalence remains high in many parts of the

Progress is evident, but stronger nutrition efforts are still needed.

Undernourishment in ASEAN Country



The percentages presented in the pie chart above illustrate the prevalence of undernourishment among the population in each country.



Depicts the severity level by country using a color scale. Green indicates safer conditions while colors ranging from orange to red indicate higher levels of undernutrition.

Factors of Stunting and Food Waste

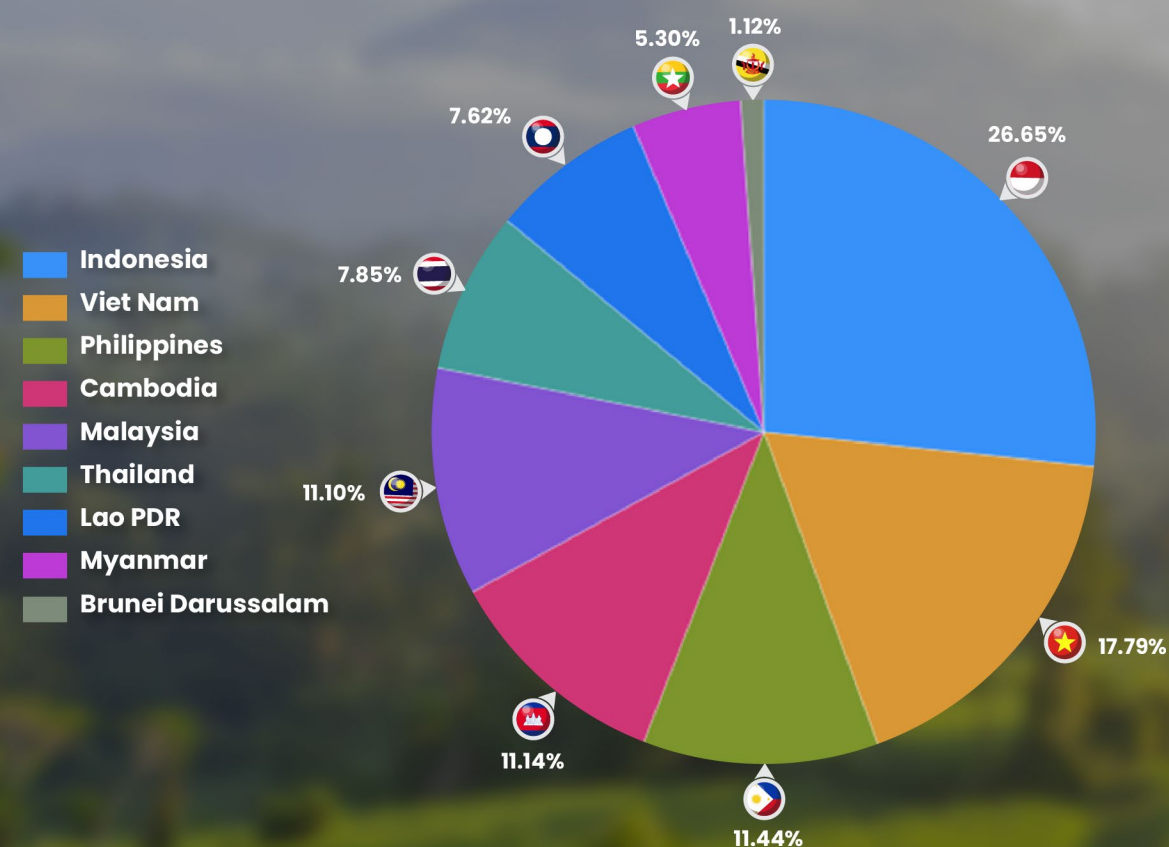


Reducing food loss and waste requires country-specific strategies. When this systemic waste is combined with poor nutrition such as a lack of access to or understanding of adequate, healthy food it creates a severe nutritional crisis



As a result, food loss, waste, and poor nutrition cause children to suffer from wasting, leaving them severely underweight and malnourished

Wasting Prevalence by Country 2016–2024



Food waste is unevenly distributed across ASEAN countries; however, several nations contribute a significantly larger share than others. Reducing food waste requires country-specific strategies.

How Wasted Food Costs Our Future Generations



1. Food Price Crisis and Food Security Emergency (SDG 2 Focus)

Inequalities within the food availability system have directly contributed to the inflation of essential food commodities, disproportionately affecting lower and middle income communities. One of the most significant examples occurred when the Philippine government officially declared a “Food Security Emergency” in early 2025. This declaration was driven by an unprecedented surge in rice prices, compelling the government to intervene through the release of national rice reserves to safeguard public access to affordable staple foods.

Source: Philippine News Agency (February 2025) – “DA declares food security emergency for rice”
Link: <https://www.pna.gov.ph/articles/1243265>



2. Persistent Threat of Chronic Malnutrition Among Younger Generations (SDG 2 Focus)

The paradox of massive food waste exists alongside an ongoing global nutrition crisis. Millions of people continue to experience limited access to nutritious and healthy food, resulting in persistent rates of childhood stunting and wasting. The latest United Nations reports highlight that these forms of malnutrition impose substantial long-term social and economic costs, demonstrating that global efforts to improve nutritional outcomes remain insufficient to achieve established sustainability targets.

Source: WHO / FAO – The State of Food Security and Nutrition in the World (July 2024)
Link: <https://www.who.int/publications/m/item/the-state-of-food-security-and-nutrition-in-the-world-2024>

How Wasted Food Costs Our Future Generations



3. Retail Sector Food Waste and Economic Inefficiency (SDG 12 Focus)

The disposal of edible food products due to commercial shelf-life limitations represents a significant source of economic inefficiency within the retail and consumption sectors. In Singapore, for example, only approximately 18% of food waste generated in 2024 was successfully recycled. This growing challenge has encouraged the emergence of innovative solutions, such as mystery food box initiatives, where restaurants and bakeries sell high-quality surplus food at discounted prices near the end of the day to reduce the accumulation of daily food waste.

Source: Channel News Asia (August 2025) –

“Mystery food boxes gaining popularity in Singapore, as merchants aim to cut food waste”

Link: <https://www.channelnewsasia.com/singapore/food-mystery-boxes-surprise-bags-gain-popularity-tackle-waste-5290316>



4. Acceleration of Climate Change and Greenhouse Gas Emissions (SDG 12 Focus)

Food waste disposed of in landfills not only undermines the sustainability of food systems but also contributes significantly to global climate change. Recent environmental assessments indicate that food loss and waste account for approximately 8–10% of global annual greenhouse gas emissions. This figure demonstrates the substantial environmental impact of food waste, with its emissions contribution estimated to be nearly five times greater than that of the entire global aviation sector.

Source: United Nations Environment Programme (UNEP) – Food Waste Index Report 2024 (March 2024)

Link: <https://www.unep.org/resources/publication/food-waste-index-report-2024>

Current Highlights in ASEAN Government

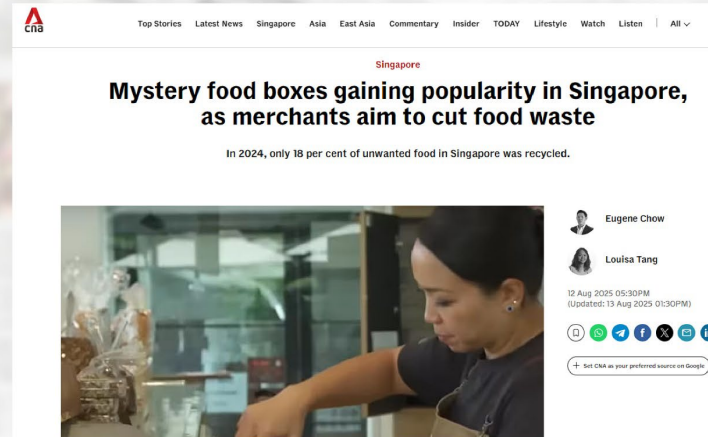


Philippines

Food Security Emergency

Emergency declared to
tame retail rice prices

Government intervention to
secure basic food supply



Singapore

Food Waste Reduction

Rise of "Mystery Food Boxes gaining
popularity

Merchants actively aiming to
cut daily food waste



Indonesia

Economic Growth & Investment

State of the Nation address by
President Prabowo

Touting significant national
investment gains

Our Solution



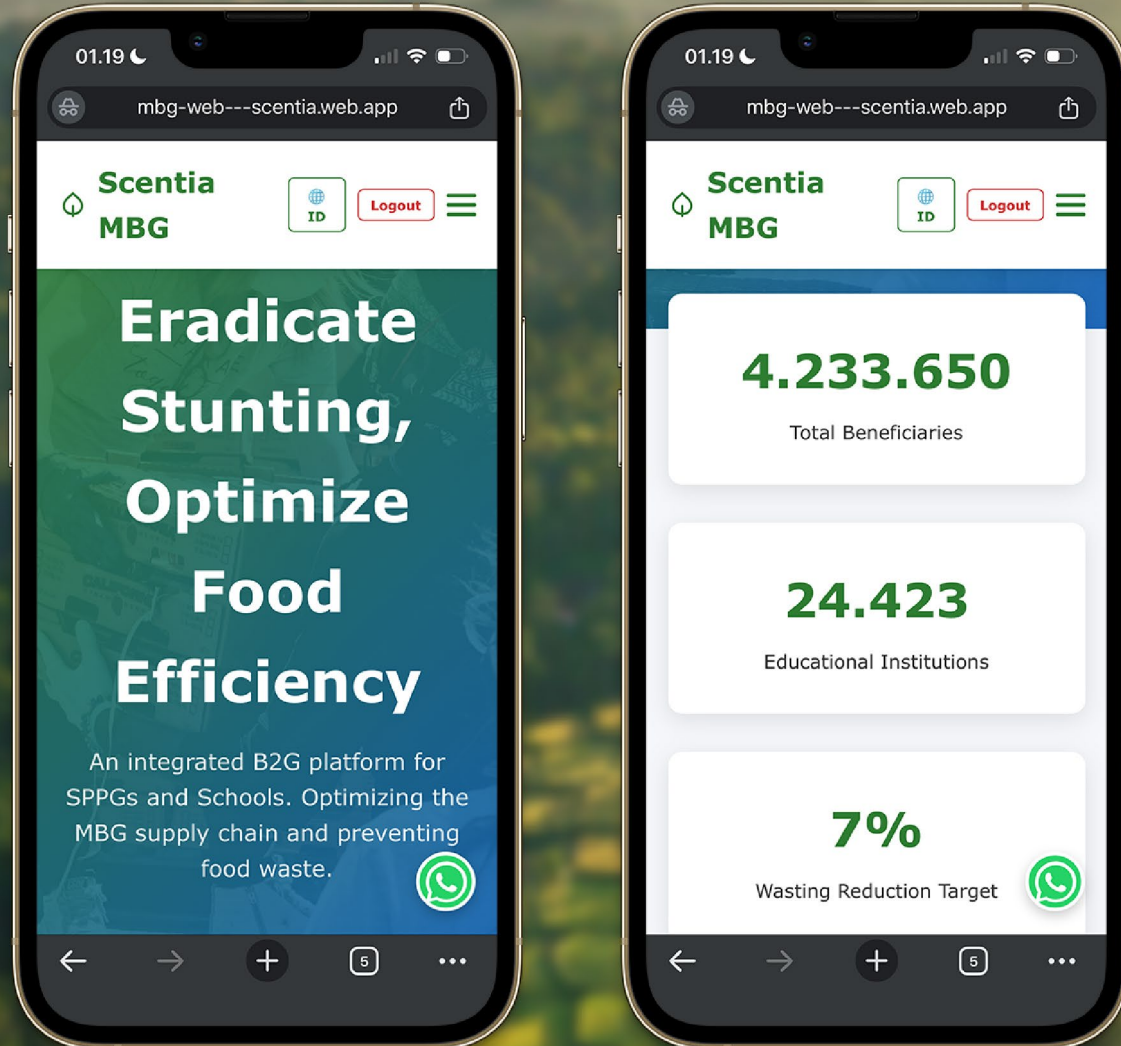
To address high stunting rates alongside massive food waste, a centralized, data driven approach is required. We therefore developed Scenia MBG, an integrated B2G web platform that directly connects Nutrition Program Service Units (SPPG) with schools. Through this website, users can monitor the equitable distribution of nutrition in real-time, track children's growth and development, and report daily leftover food portions. This digital system enables the government to optimize the Free Nutritious Meal supply chain, ensuring greater targeting precision and successfully achieving the goal of reducing

OUR SOLUTION



SCAN HERE!

Home Page



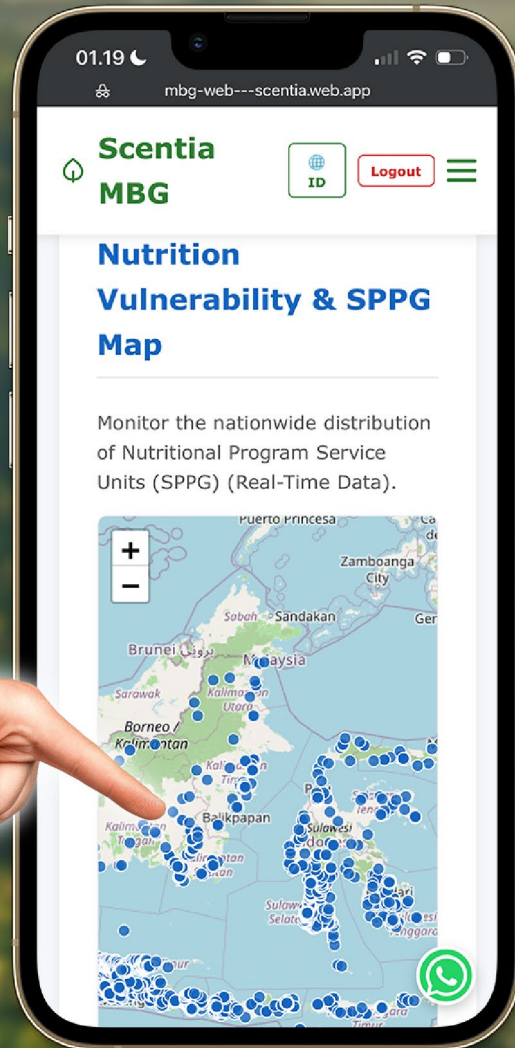
This Scentia MBG homepage showcases the prototype of a B2G digital platform designed by the Scentia Collective team for the ASEAN Data Science Explorers competition.

Conceptually, this integrated platform is designed to connect nutrition service units (SPPG) with educational institutions to combat stunting and optimize food supply chain efficiency.

The statistics displayed on the dashboard specifically the 4,233,650 total beneficiaries and 24,423 educational institutions are real data sourced directly from the National Nutrition Agency (BGN) website, representing the program's target scale in the field.

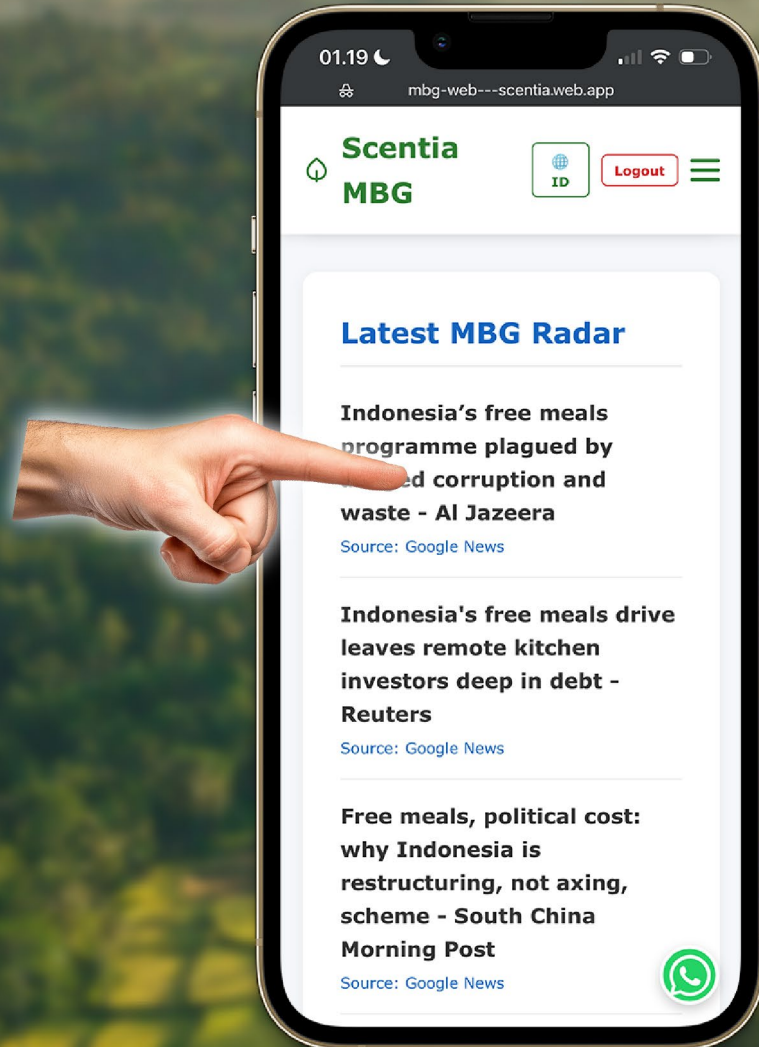
By implementing the system's features at this scale, we aim to optimize the distribution of nutritious meals and reduce food waste by 7%.

Home Page



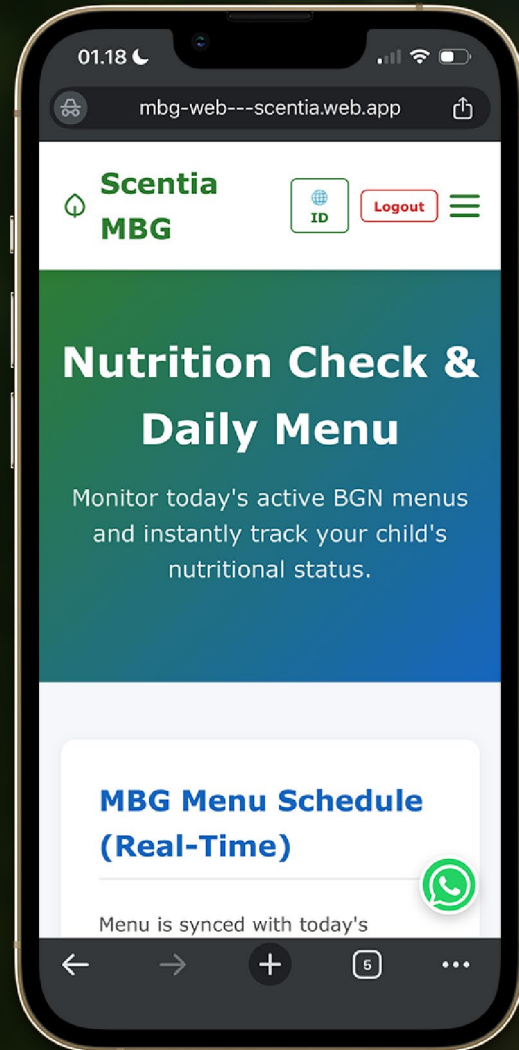
This interactive map visualizes the real-time nationwide distribution of Nutritional Program Service Units (SPPG) using blue dots to monitor equitable program coverage.

Home Page



This capability automatically gathers the most recent information and reports related to the MBG project to monitor public opinion and challenges faced in the project.

Nutrition Check Page



This page includes not only daily menu planning but also the option of instant checking of the child's nutritional status through the calorie calculator.

Nutrition Check Page



• MBG Menu Assessment •

MBG Menu Schedule Real Time will present the menu schedule that has been coordinated with the information obtained from the National Nutrition Agency in real time every day. For instance, when a person eats white rice, crispy fried fish, vegetable lodeh, and soy milk, the total calorie intake will be 460 kcal. As the nature of each type of food differs regarding its nutritional value and calories, the energy intake will always be calculated precisely depending on the variations of the menu served.

Nutrition Check Page

01:18

mbg-web---scentia.web.app

Scentia
MBG

ID

Logout

Child Growth Check

Enter the child's current weight and height to instantly monitor their nutritional status.

Child Age (months)

0-60

Gender

- Select -

Weight (kg)

Example: 12.5

Height / Length (cm)

Example: 85.0

Check Nutrition Status
Now

Child Growth Check



Instant Nutrition Status

Recording data on age, gender, weight, and height to instantly assess the child's nutritional status.



Comprehensive Metric Input

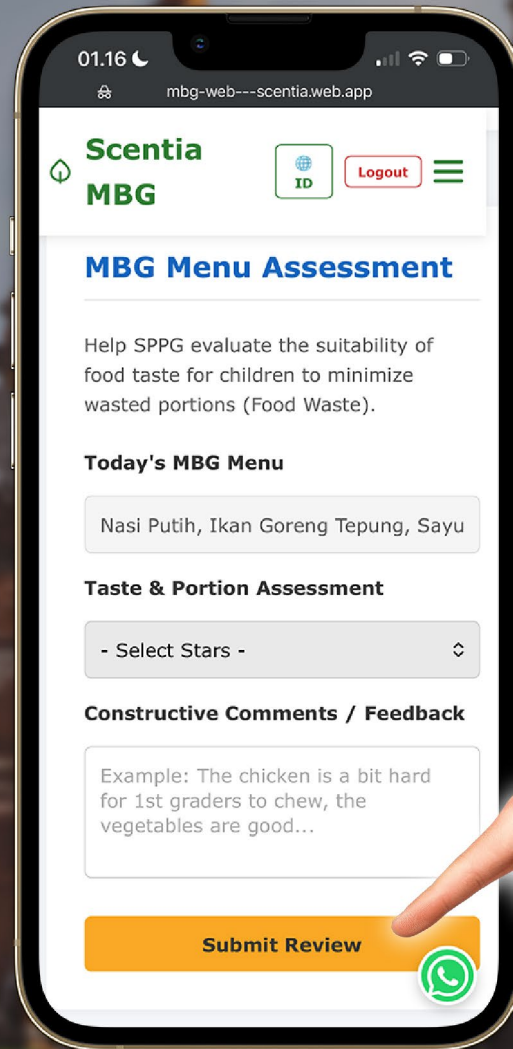
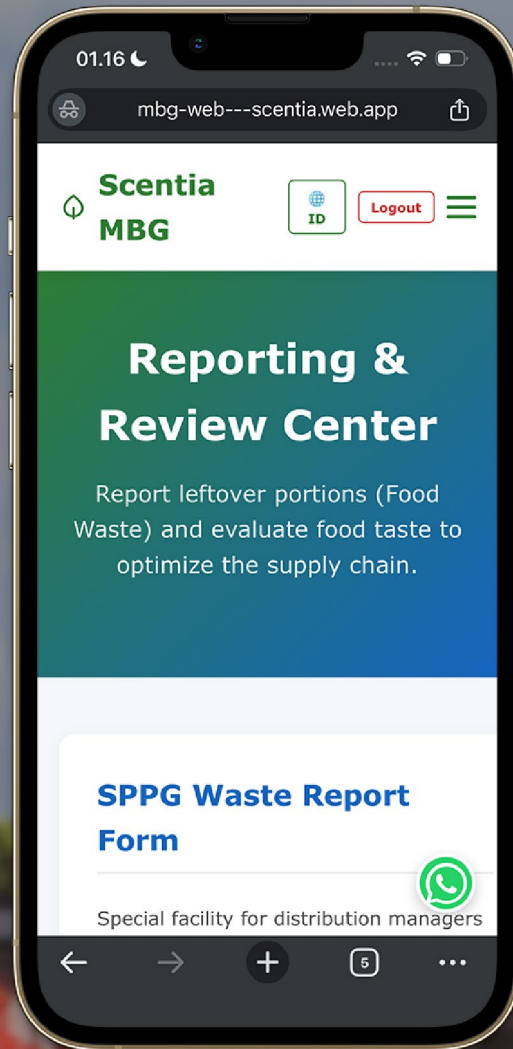
Displays the nutritional status evaluation result, complete with the calculated score.



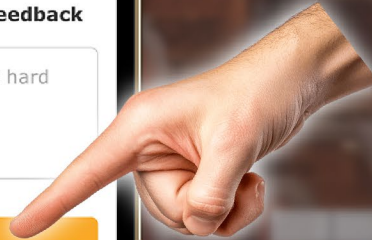
Actionable Health

Providing health education aligned with the government's MBG program, such as reminders about the importance of washing hands with soap.

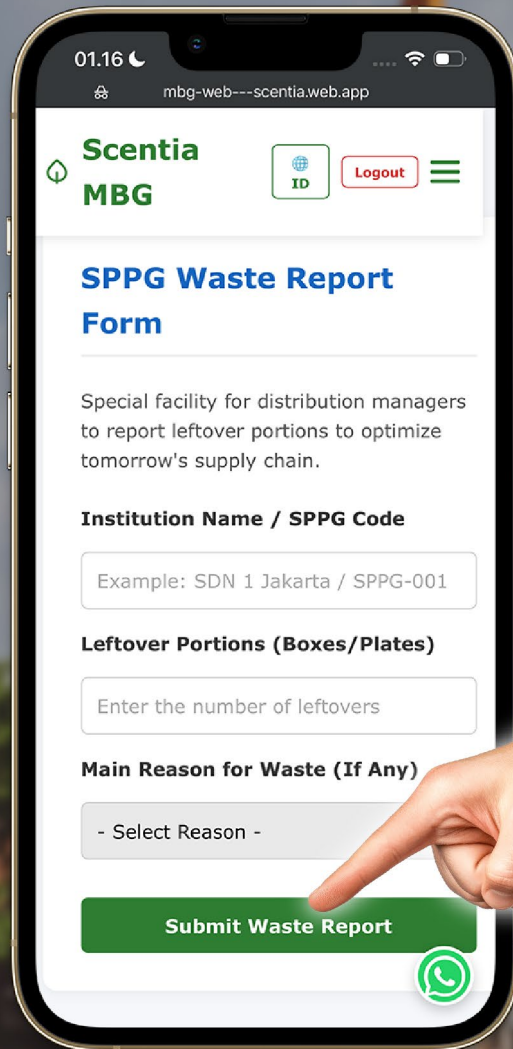
Report



This feature allows users to submit ratings and constructive feedback regarding the daily menu's taste and portion sizes to help evaluate suitability and minimize food waste.



Report



01.16 🌙
mbg-web---scentia.web.app

Scentia MBG ID Logout

SPPG Waste Report Form

Special facility for distribution managers to report leftover portions to optimize tomorrow's supply chain.

Institution Name / SPPG Code

Example: SDN 1 Jakarta / SPPG-001

Leftover Portions (Boxes/Plates)

Enter the number of leftovers

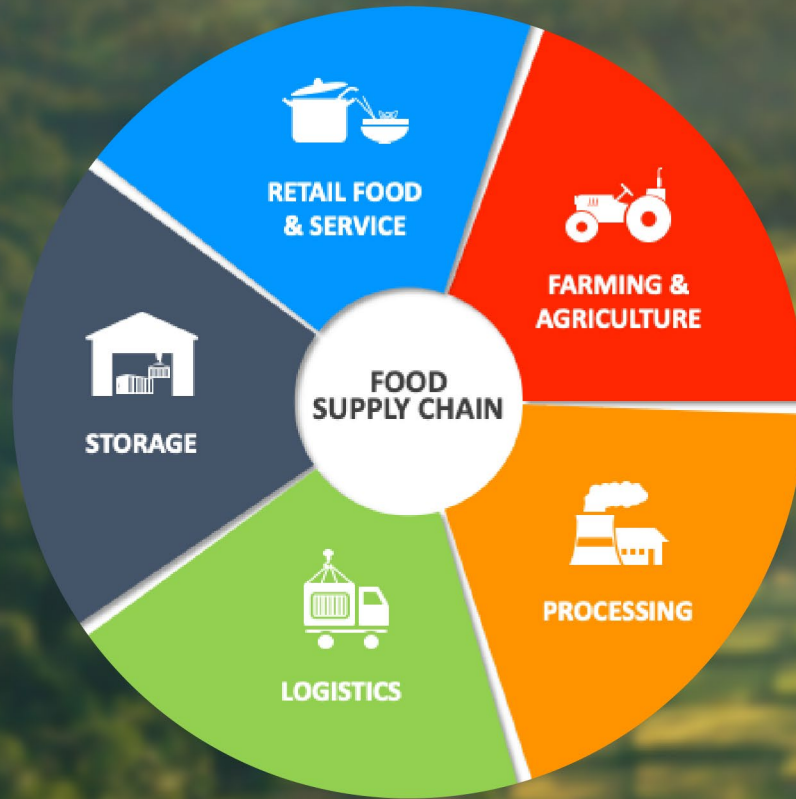
Main Reason for Waste (If Any)

- Select Reason -

Submit Waste Report

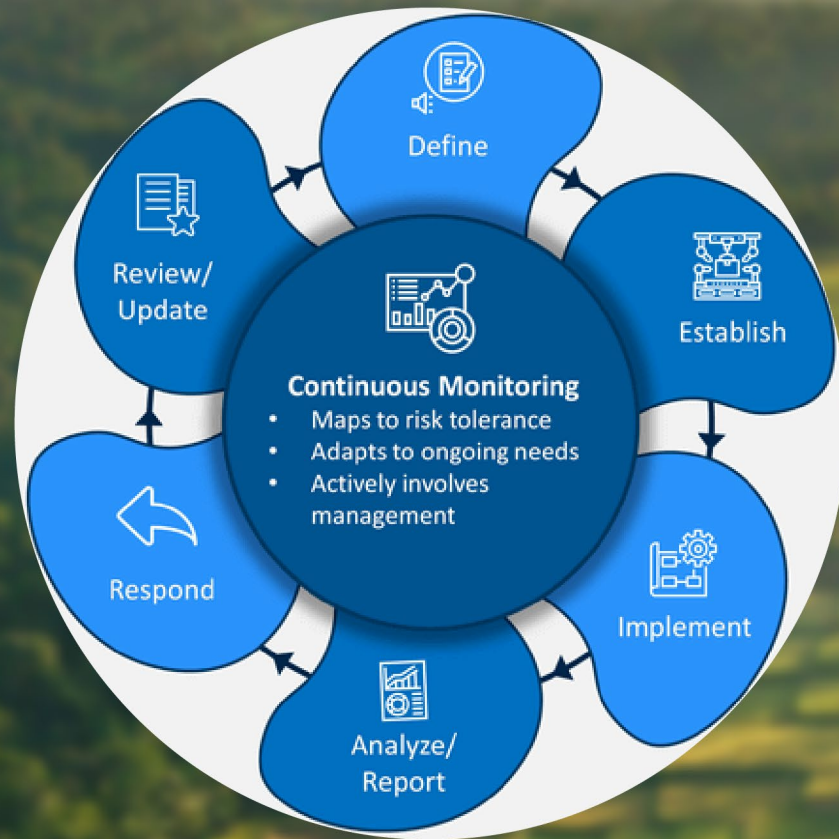
WhatsApp icon

This Page allows users to report leftover food and evaluate daily menu taste to help optimize the overall MBG supply chain.



Supply Chain Optimization and Food Waste Reduction

This website plays an active role in reducing food wastage by targeting a reduction in food wastage of 7%. The SPPG food wastage report form along with the menu assessment, helps the distribution managers record any leftover food as well as the appropriateness of its taste in relation to children. This information directly impacts the optimization of the supply chain of the MBG program, which ensures that the next day's supply is efficient and precise.



Monitoring Growth, Development and Stunting Prevention

The website contributes to the eradication of stunting directly through its nutrition assessment tools, which are made easy to access. The "Child Growth Check" tool provides for immediate measurement of the nutritional status of a child based on physical attributes including age, sex, weight, and height. Furthermore, the real-time synchronized meal plan, which has the estimated calorie count, ensures that the nutritional intake of the child is



Supporting the Government's Food Security Agenda

Various ASEAN nations are currently addressing food security challenges while prioritizing economic growth including Indonesia, under President Prabowo's national investment agenda. Scenia MBG fully supports this strategic initiative by providing a Nutrition Vulnerability Map that transparently charts nutritional vulnerability and distribution points, ensuring that government investments in the Free Nutritious Meal (MBG) program are effectively targeted and yield maximum impact on the human capital of the future.

How Scientia MBG Aligns with ASEAN Priorities & **SDGs**



ASEAN Socio-Cultural & SDG Alignment

Strategic Goal

How Our App Contributes



SDG 2: Zero Hunger & Nutrition Security

Actively tracks Free Nutritious Meal (MBG) distribution and provides an instant Child Growth Check tool to assess nutritional status and combat stunting.



Inclusive Access to Nutriton Data

Empowers communities with real-time access to daily meal schedules and calorie estimations directly synchronized with the National Nutrition Agency.



Digital Transparency in Food Distribution

Deploys an interactive Nutrition Vulnerability & SPPG Map to visualize service unit locations, ensuring equitable program coverage nationwide.



Community-Driven Food Quality Control

Utilizes the MBG Menu Assessment feature to gather user feedback on meal taste and suitability for children, directly mitigating potential food waste.



ASEAN Economic & Digital Alignment

Strategic Goal

How Our App Contributes



SDG 12: Responsible Consumption & Production

Commits to a targeted 7% reduction in food waste by rigorously monitoring and evaluating leftover portions within the MBG ecosystem.



Innovation-Driven Supply Chain Optimization

Equips distribution managers with the SPPG Waste Report Form to log unconsumed meals, ensuring next-day food logistics are accurate and resource-efficient.



Data-Driven Food Security Investments

Provides actionable, real-time data to help the government target nutritional investments precisely, minimizing economic inefficiencies.



Circular Economy Integration

Strategically unites the mission of eradicating hunger (SDG 2) with sustainable food management (SDG 12) to maximize resource efficiency.





Conclusion

The presence of childhood stunting and malnutrition among ASEAN countries has been seriously undermined by the existence of extensive food loss and inefficiencies in the food chain industry. With regard to tackling this problem, Scentia MBG has developed a complete B2G digital platform that enhances the Free Nutritious Meal (MBG) supply chain through tracking and direct evaluation. Using the platforms of child growth monitoring and food reporting center, the program will actively reduce the loss of food and ensure that the meals contain dietary nutrients according to their tastes. In doing so, the program will fulfill national food security plans through reducing food waste by

A scenic view of a traditional Chinese temple complex. In the foreground, a tall, multi-tiered pagoda stands prominently. To its right, a smaller pavilion with a traditional roof is visible. The entire scene is reflected in a calm body of water. In the background, misty mountains are visible under a soft, hazy sky. The overall atmosphere is peaceful and serene.

THANK YOU

References

- [1] Food and Agriculture Organization. (n.d.). Suite of Food Security Indicators [Data set]. FAOSTAT. Retrieved from <https://www.fao.org/faostat/en/#data/FS>
- [2] World Bank. (n.d.). Prevalence of wasting, weight for height (% of children under 5) (SH.STA.WAST.ZS) [Data set]. World Development Indicators. Retrieved from <https://data.worldbank.org/indicator/SH.STA.WAST.ZS>
- [3] World Bank. (n.d.). Prevalence of stunting, height for age (% of children under 5) (SH.STA.STNT.ZS) [Data set]. World Development Indicators. Retrieved from <https://data.worldbank.org/indicator/SH.STA.STNT.ZS>
- [4] World Bank. (n.d.). Urban population (SP.URB.TOTL) [Data set]. World Development Indicators. Retrieved from <https://data.worldbank.org/indicator/SP.URB.TOTL>
- [5] World Bank. (n.d.). Food production index (2014-2016 = 100) (AG.PRD.FOOD.XD) [Data set]. World Development Indicators. Retrieved from <https://data.worldbank.org/indicator/AG.PRD.FOOD.XD>
- [6] World Bank. (n.d.). Birth rate, crude (per 1,000 people) - Low income (SP.DYN.CBRT.IN) [Data set]. World Development Indicators. Retrieved from <https://data.worldbank.org/indicator/SP.DYN.CBRT.IN?locations=XM>
- [7] World Bank. (n.d.). Population growth (annual %) (SP.POP.GROW) [Data set]. World Development Indicators. Retrieved from <https://data.worldbank.org/indicator/SP.POP.GROW>

Appendix [1]

Urban Population (Total) in ASEAN Countries (2016-2024)

No.	Countries	Urban Population (Annual Count)								
		2016	2017	2018	2019	2020	2021	2022	2023	2024
1	Brunei Darussalam	330,167	332,567	337,167	339,467	344,067	346,367	348,667		
2	Cambodia	5,148,474	5,567,774	6,125,801	6,652,399	6,808,856	6,945,450	7,209,455		
3	Indonesia	142,402,781	145,581,816	151,880,690	154,667,731	160,309,381	169,309,381	172,551,891		
4	Lao PDR	2,337,023	2,337,023	2,259,022	2,269,900	2,877,932	2,974,555	3,073,369		
5	Myanmar	15,352,369	15,352,369	15,677,833	15,833,415	16,144,275	16,164,969	16,595,228		
6	Malaysia	23,479,511	23,479,511	23,439,517	25,026,617	25,886,197	26,655,901	27,349,906		
7	Philippines	55,408,086	55,408,086	52,425,856	05,519,606	60,444,331	62,431,327	64,237,746		
8	Singapore	5,607,283	5,607,283	5,683,763	5,685,807	5,553,026	6,597,020	6,036,860		
9	Thailand	33,927,770	33,927,770	38,903,010	39,999,996	41,046,616	44,369,130	44,340,234		
10	Timor-Leste	361,692	361,692	377,692	384,050	391,422	395,840	401,480		
11	Viet Nam	31,876,800	32,153,511	32,710,364	32,986,736	33,544,944	33,823,181	34,101,231		

Appendix [2]

Population Growth Rate (%) in ASEAN Countries (2016-2024)

No.	Countries	Annual Population Growth Rate (%)								
		2016	2017	2018	2019	2020	2021	2022	2023	2024
1	Brunei Darussalam	1.29	1.25	1.30	1.28	1.31	1.29	1.34	1.32	1.35
2	Cambodia	0.81	1.02	0.90	0.66	<u>1.52</u>	0.79	0.82	0.98	0.53
3	Indonesia	0.94	0.99	0.99	0.80	1.05	0.81	0.97	1.05	0.98
4	Lao PDR	0.83	0.90	0.79	0.95	1.07	1.08	0.92	0.87	1.03
5	Myanmar	0.87	0.84	0.83	0.97	1.04	0.80	0.96	0.60	0.81
6	Malaysia	0.76	0.57	0.73	0.01	1.10	0.57	1.03	0.66	1.02
7	Philippines	0.71	0.69	0.91	0.31	1.21	0.48	0.93	0.70	0.94
8	Singapore	0.79	0.62	0.31	0.08	1.02	0.45	1.09	0.52	0.74
9	Thailand	0.83	0.54	0.31	0.13	0.70	0.42	1.19	0.47	0.83
10	Timor-Leste	0.77	0.22	0.17	0.01	0.86	0.24	0.97	0.25	0.97
11	Viet Nam	0.83	0.45	0.41	0.30	0.59	0.46	1.05	0.37	1.03

Appendix [3]

Birth Rate in ASEAN Countries (2016-2024)

No.	Countries	Birth Rate (per 1000 people)								
		2016	2017	2018	2019	2020	2021	2022	2023	2024
1	Brunei Darussalam	25.8	26.7	27.0	27.3	24.0	26.9	25.8	28.7	24.5
2	Cambodia	18.9	19.8	16.1	18.6	20.4	24.7	20.9	20.9	26.3
3	Indonesia	21.7	21.4	22.0	20.8	22.2	20.1	21.7	21.3	22.0
4	Lao PDR	19.7	15.8	16.2	19.5	16.8	17.8	18.9	18.7	10.3
5	Myanmar	18.7	14.0	15.4	19.7	13.1	14.9	15.7	16.6	18.1
6	Malaysia	17.6	15.4	13.3	0.1	12.0	5.5	14.3	10.7	10.2
7	Philippines	17.7	16.9	19.1	1.2	12.1	4.8	10.3	7.0	19.4
8	Singapore	9.1	9.1	7.4	0.8	8.2	8.4	9.1	8.0	8.0
9	Thailand	0.83	5.4	3.1	0.3	7.0	4.2	11.9	4.7	0.83
10	Timor-Leste	29.5	02.2	11.7	01.7	08.6	02.4	09.7	02.5	26.1
11	Viet Nam	0.83	0.45	0.41	0.30	0.59	0.46	1.05	0.37	1.05